Math 1151 — Midterm 4 (Excerpt)

Problem 1 *A function h is continuous on its domain $[-4, 6]$. The graph of $h'(x)$ is given.*

- *List all critical points.*
- *List all inflection points.*

Problem 2 *A function f and its derivative f' have domain $(-\infty, \infty)$. Some values are given below:*

x	-1	0	1	2
$f(x)$	-12	-9	-7	-5
$f'(x)$	5	3	$\frac{3}{2}$	$\frac{1}{2}$

(a) *Find the linear approximation $L(x)$ for f at $x = 1$.*

(b) *Use $L(x)$ to estimate $f(1.3)$.*

(c) *Explain whether your estimate is an overestimate or an underestimate.*

Problem 3 *Use the properties of logarithms to write the expression as a single logarithm.*

- $4\ln(x) + 2\ln(y) = \ln\left(x^4y^2\right)$
- $2\ln(x) - 3\ln(y) - 4\ln(z) = \ln\left(\frac{x^2}{y^3z^4}\right)$
- $\ln(x) - 3\ln(z) + \frac{1}{2}\ln(y) = \ln\left(\frac{x\sqrt{y}}{z^3}\right)$
- $-3\ln(x) - 3\ln(y) + 3\ln(z) = \ln\left(\frac{z^3}{x^3y^3}\right)$
- $\ln(x) + \frac{1}{2} = \ln\left(e^{1/2}x\right)$

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